

Question 1.

Solve the following systems of linear equations by Gaussian elimination method:

(i) $2x - 2y + 3z = 2$, $x + 2y - z = 3$, $3x - y + 2z = 1$

$$2x - 2y + 3z = 2$$

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$A = \left(\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array} \right)$$

Swap R2 $\longleftrightarrow$ R1

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array} \right)$$

Perform R2 $\rightarrow$ R2 - 2R1

Perform R3 $\rightarrow$ R3 - 3R1

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{array}\right)$$

$$C2 \leftrightarrow C3$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 5 & -6 & -4 \\ 0 & 5 & -7 & -8 \end{array}\right)$$

$$R3 \rightarrow R3 - R2$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 5 & -6 & -4 \\ 0 & 0 & -1 & -4 \end{array}\right)$$

Solve for z:

$$-z = -4$$

$$z = 4$$

Solve for y:

$$5y - 6z = -4$$

$$5y - 6(4) = -4$$

$$5y - 24 = -4$$

$$5y = 20$$

$$y = 4$$

Solve for x:

$$x - y + 2z = 3$$

$$x - 4 + 2(4) = 3$$

$$x - 4 + 8 = 3$$

$$x + 4 = 3$$

$$x = -1$$

Final Answer:

$$x = -1$$

$$y = 4$$

$$z = 4$$

(ii) $2x + 4y + 6z = 22$, $3x + 8y + 5z = 27$, $-x + y + 2z = 2$

$$2x + 4y + 6z = 22$$

$$3x + 8y + 5z = 27$$

$$-x + y + 2z = 2$$

$$A = \left(\begin{array}{ccc|c} 2 & 4 & 6 & 22 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right)$$

$$R1 \rightarrow R1/2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right)$$

$$\text{Perform } R2 \rightarrow R2 - 3R1$$

$$\text{Perform } R3 \rightarrow R3 + R1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 2 & -4 & -6 \\ 0 & 3 & 5 & 13 \end{array} \right)$$

$$R2 \rightarrow R2/2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 3 & 5 & 13 \end{array} \right)$$

$$R3 \rightarrow R3 - 3R2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 11 & 22 \end{array} \right)$$

Solve for z:

$$11z = 22$$

$$z = 2$$

Solve for y:

$$y - 2z = -3$$

$$y - 2(2) = -3$$

$$y - 4 = -3$$

$$y = 1$$

Solve for x:

$$x + 2y + 3z = 11$$

$$x + 2(1) + 3(2) = 11$$

$$x + 2 + 6 = 11$$

$$x + 8 = 11$$

$$x = 3$$

Final Solution:

$$x = 3$$

$$y = 1$$

$$z = 2$$

Question 2.

If $ax^2 + bx + c$ is divided by $x + 3$, $x - 5$, and $x - 1$, the remainders are 21, 61 and 9 respectively. Find a , b and c . (Use Gaussian elimination method.)

$$P(x) = ax^2 + bx + c$$

$$P(-3) = 21$$

$$P(5) = 61$$

$$P(1) = 9$$

$$9a - 3b + c = 21$$

$$25a + 5b + c = 61$$

$$a + b + c = 9$$

$$A = \left(\begin{array}{ccc|c} 9 & -3 & 1 & 21 \\ 25 & 5 & 1 & 61 \\ 1 & 1 & 1 & 9 \end{array} \right)$$

Swap $C1 \leftrightarrow C3$

$$\left(\begin{array}{ccc|c} 1 & -3 & 9 & 21 \\ 1 & 5 & 25 & 61 \\ 1 & 1 & 1 & 9 \end{array}\right)$$

Perform $R_2 \rightarrow R_2 - R_1$

Perform $R_3 \rightarrow R_3 - R_1$

$$\left(\begin{array}{ccc|c} 1 & -3 & 9 & 21 \\ 0 & 8 & 16 & 40 \\ 0 & 4 & -8 & -12 \end{array}\right)$$

$R_2 \rightarrow R_2 / 8$

$R_3 \rightarrow R_3 / 4$

$$\left(\begin{array}{ccc|c} 1 & -3 & 9 & 21 \\ 0 & 1 & 2 & 5 \\ 0 & 1 & -2 & -3 \end{array}\right)$$

$R_3 \rightarrow R_3 - R_2$

$$\left(\begin{array}{ccc|c} 1 & -3 & 9 & 21 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & -4 & -8 \end{array}\right)$$

Solve for c

$$-4c = -8$$

$$c = 2$$

Solve for b

$$b + 2c = 5$$

$$b + 2(2) = 5$$

$$b = 5 - 4$$

$$b = 1$$

Solve for a

$$a - 3b + 9c = 21$$

$$a - 3(1) + 9(2) = 21$$

$$a - 3 + 18 = 21$$

$$a + 15 = 21$$

$$a = 6$$

Final Result:

The polynomial is $P(x) = 6x^2 + x + 2$.

Question 3.

An amount of ₹65,000 is invested in three bonds at the rates of 6%, 8% and 9% per annum respectively. The total annual income is ₹4,800. The income from the third bond is ₹600 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method.)

Solution:

Amount invested in bond I = x

Amount invested in bond II = y

Amount invested in bond III = z

$$x + y + z = 65,000 \text{} \rightarrow (1)$$

$$\frac{6x}{100} + \frac{8y}{100} + \frac{9z}{100} = 4800$$

$$6x + 8y + 9z = 480000 \text{} \rightarrow (2)$$

$$9 \% \text{ of } z = 8 \% \text{ of } y + 600$$

$$\frac{9z}{100} - \frac{8y}{100} = 600$$

$$9z - 8y = 60000 \quad \cdots \longrightarrow (3)$$

Augmented matrix

$$(A|B) = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 6 & 8 & 9 & 480000 \\ 0 & -8 & 9 & 60000 \end{array} \right]$$

$$R2 \longrightarrow R3 - 6R1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 3 & 90000 \\ 0 & -8 & 9 & 60000 \end{array} \right]$$

$$R3 \longrightarrow R3 + 4R2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 3 & 90000 \\ 0 & 0 & 21 & 420000 \end{array} \right]$$

$$x + y + z = 65000 \quad \cdots \rightarrow (4)$$

$$2y + 3z = 90000 \quad \cdots \rightarrow (5)$$

$$21z = 420000 \quad \cdots \rightarrow (6)$$

From (6) :

$$21z = 420000$$

$$z = 20000$$

Sub z in (5):

$$2y + 3(20000) = 90000$$

$$2y = 90000 - 60000$$

$$2y = 30000$$

$$y = 15000$$

Sub z and y in (4):

$$x + 15000 + 20000 = 65000$$

$$x + 35000 = 65000$$

$$x = 30000$$

Final Answer:

$$x = 30000 ; y = 15000 ; z = 200$$

Question 4.

A boy is walking along the path $y = ax^2 + bx + c$ through the points $(-6, -8)$, $(-2, -12)$, and $(3, 8)$. He wants to meet his friend at $P(7, 60)$. Will he meet his friend? (Use Gaussian elimination method.)

Solution:

$$y = ax^2 + bx + c \quad \text{.....} \rightarrow (1)$$

point 1: $(-6, -8)$

$$a(-6)^2 + b(-6) + c = 8$$

$$36a - 6b + c = 8 \quad \text{.....} \rightarrow (2)$$

point 2: $(-2, -12)$

$$4a - 2b + c = 12 \quad \text{.....} \rightarrow (3)$$

point 3: (3,8)

$$9a + 3b + c = 8 \quad \cdots \longrightarrow (4)$$

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 4 & -2 & 1 & -12 \\ 9 & 3 & 1 & 8 \end{array} \right] \begin{array}{l} \\ R2 \longrightarrow R2*9 \\ R3 \longrightarrow R3*4 \end{array}$$
$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 36 & -18 & 9 & -108 \\ 36 & 12 & 4 & 32 \end{array} \right] \begin{array}{l} \\ R2 \longrightarrow R2 - R1 \\ R3 \longrightarrow R3 - R1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -12 & 8 & -116 \\ 0 & 18 & 3 & 24 \end{array} \right] \quad R2 \longrightarrow R2/2$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -6 & 4 & -58 \\ 0 & 18 & 3 & 24 \end{array} \right] \quad R3 \longrightarrow R3 + 3R2$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -6 & 4 & -58 \\ 0 & 0 & 15 & -150 \end{array} \right]$$

$$36a - 6b - c = 8 \quad \text{.....} \rightarrow (5)$$

$$-6b + 4c = -58 \quad \text{.....} \rightarrow (6)$$

$$15c = -150 \quad \text{.....} \rightarrow (7)$$

From (7)

$$c = -10$$

Sub c in (6)

$$-6b - 40 = -58$$

$$-6b = -18$$

$$b = 3$$

Sub b and c in (5)

$$36a - 18 - 10 = 8$$

$$36a = 36$$

$$a = 1$$

Sub a, b, and c in (1)

$$y = x^2 + 3x - 10$$

Sub the point $x = 7$

$$y = 49 + 3(7) - 10$$

$$= 49 + 21 - 10$$

$$= 70 - 10$$

$$y = 60$$

Yes, he will meet his friend at (7,60)